HGL User Guide

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Safety Practices

This instruction manual applies only to HG-Series LOW-VOLTAGE AC POWER CIRCUIT BREAKER (ACB) regarding installation and maintenance procedures. Installing and maintaining these products improperly may result in serious personal injury, property damage, or even death. Therefore this instruction manual must be read and understood at any step in unpacking, assembly, operation, and maintenance of the breaker. Only qualified persons who are familiar with installing and maintaining circuti breakers are permitted to work on breakers, and this instruction manual should be accessible to those persons at any time.
Please refer to Detail Instruction Manual by the QR Code.

Transportation Precaution

Never lift the ACB over an area where there are people.
Never stand under the ACB.

OCR Field Test Precautions

- OCR field tests and setting changes must be performed by qualified personnel.
- After completion of OCR tests, make sure to return the original values.
- Failure to do may cause a fire or burnout.

Operation Precautions

- Never touch live terminal parts.
  There is a risk of electric shock.
- Do not leave the ACB in the draw-out position.
  Dropping the ACB cause serious injury.
- If the ACB trips open automatically, remove the cause before closing the ACB.
  Otherwise, a fire could occur.
- For the ACB with the fixing block, be sure to loosen the block screws before drawing out the ACB.
  Otherwise, damage to the ACB may occur.

Installation Precautions

- Installation work must be performed by qualified personnel.
- Prior to commencing any installation work, open an upstream circuit breaker to shut off all sources of power.
  Otherwise, electric shock may occur.
- Tighten terminal bolt up securely to the specified torque.
  Otherwise, a fire may occur.
- Fix the draw-out cradle of the ACB firmly on a flat level surface using mounting screws. **Otherwise, draw-out operation may cause the ACB to fall.**
- To ensure adequate arc space (insulation distance), avoid blocking the arc gas vents of the ACB.
  Otherwise a burn may result from high temperature arc gas.
- Do not place the ACB in such as an area that is subject to high temperature, high humidity, dusty air, corrosive gas, vibration and shock, or other unusual conditions. **Mounting in such area could cause a fire, non-tripping, or malfunction.**
- Install the ACB to prevent dust, concrete powder, iron powder,a nd rainwater from entering the unit.
  These materials could cause a fire or non-tripping.
- For the ACB with four poles, connect the neutral wire of a 3-phase, 4-wire cable to the N-phase pole(on the right end).
  Otherwise, an overcurrent may cause non-tripping resulting a fire.

Maintenance and Inspection Precautions

- ACB maintenance, inspection and/or parts replacement must be performed by qualified personnel.
- Prior to commencing any work on the ACB, open an upstream breaker or the like to isolate all sources of power/voltage from both the primary anda uxiliary circuits.
  Otherwise, electric shock may result.
- Prior to commencing ACB internal inspection, make sure that the ACB is open and the closing spring is released. Otherwise, fingers or tools could be pinched in the **internal mechanism, resulting injury.**
- Retighten the terminal screws periodically to the specified torque.
  Otherwise, a fire could result.
- Be sure to reinstall the arc chamber if removed.
  Failure to do so or incorrect installation of the arc chamber may result in a fire or burn.
- Do not touch ACB live parts (contacts in particular), or structural parts close to a live part immediately after opening the ACB to stop supplying power.
  Otherwise, remaining heat may cause burn or residual charged voltage may cause electric shock.
- Do not bring your hand or face close to the arc gas vent of th earc chute while the ACB is closed. **Otherwise, a burn may result from high-temperature arc gas blowing out of the arc gas vent when the ACB trips open.**

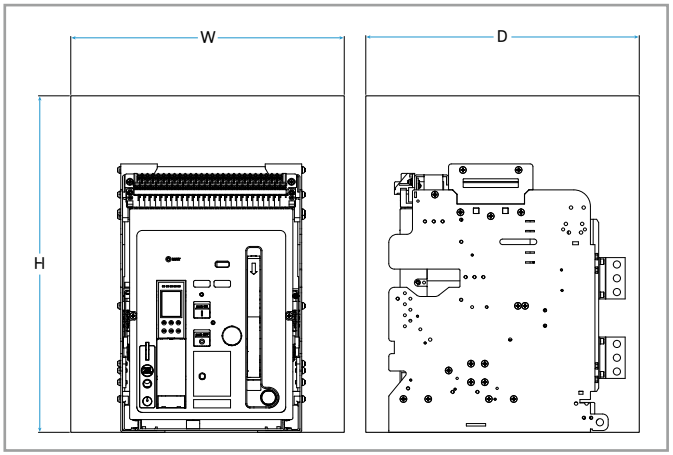
Table1. Minimum enclosure size

Enclosure Dimension			H	W	D
A Frame	3P	In	19.69	21.18	15.00
		mm	500	538	381
	4P	In	19.69	24.53	15.00
		mm	500	623	381
B&Q Frame	3P	In	19.69	24.02	15.00
		mm	500	610	381
	4P	In	19.69	28.50	15.00
		mm	500	725	381
C Frame	3P	In	20.47	32.83	15.00
		mm	520	834	381
	4P	In	20.47	39.53	15.00
		mm	520	1004	381
D Frame	3P	In	20.47	38.43	15.00
		mm	520	976	381
	4P	In	20.47	47.48	15.00
		mm	520	1206	381

※ 1) Enclosure dimension is followed by UL/ANSI Standard.

Table2. Minimum enclosure ventilation

Enclosure ventilation		Top mm²	Bottom mm²
A Frame	3P	12,000	12,000
	4P	15,000	15,000
B&Q Frame	3P	35,000	35,000
	4P	45,000	45,000
C Frame	3P	50,000	50,000
	4P	50,000	50,000
D Frame	3P	50,000	50,000
	4P	50,000	50,000



Draw-out Mechanism

In order to test the breaker and exchange parts easily, the breaker body is drawn-in /out from draw-out cradle, the breaker can be fixed in one among three of draw-out cradle. The breaker can be moved to TEST or ISOLATED position when the panel door is closed.

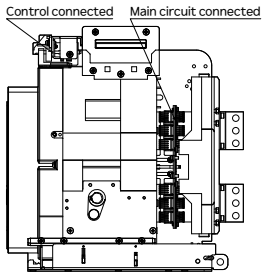
Caution of operation

- Open the breaker before working with the draw-out mechanism.
- Loosen the screws of the fixing blocks before drawing out the breaker body.
- Push the position lock release button, insert draw-in/out handle into the insertion hole.
- When you try to move the breaker body to the CONNECTED position, the operating force will increase.
  Torque of operating force is about 25 kgf.
- Operate draw-in/out handle totally in inserted state.
- Rotate draw-in/out handle until the handle is locked automatically at each ISOLATED, TEST, and CONNECTED position.
  At each locked position, stop turning the draw-out handle.
  Failure to follow this instruction can result in equiptment damage or malfunction.

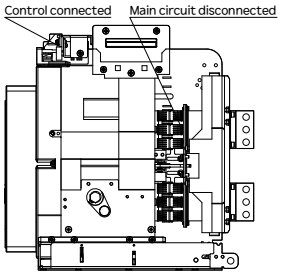
CAUTION

During the breaker body move from ISOLATED to CONNECTED position, or from CONNECTED to ISOLATED position, do not rotate the handle by releasing position locking.

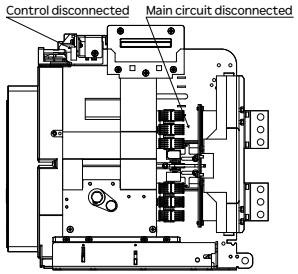
CONNECTED Position



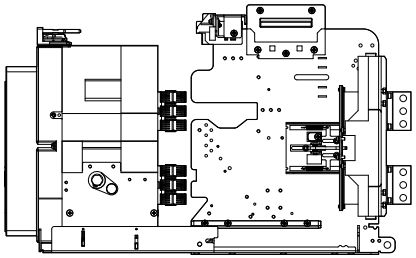
TEST Position



ISOLATED Position



REMOVED Position



Draw-out Operation

The draw-in/out handle is used to move the breaker body to oneo f the three positions (CONNECTED, TEST, ISOLATED).

! CAUTION

When the fixing blocks (Fig. 1. ② option) are fitted, loosen the right and left screws of the fixing blocks before drawing-out operation.

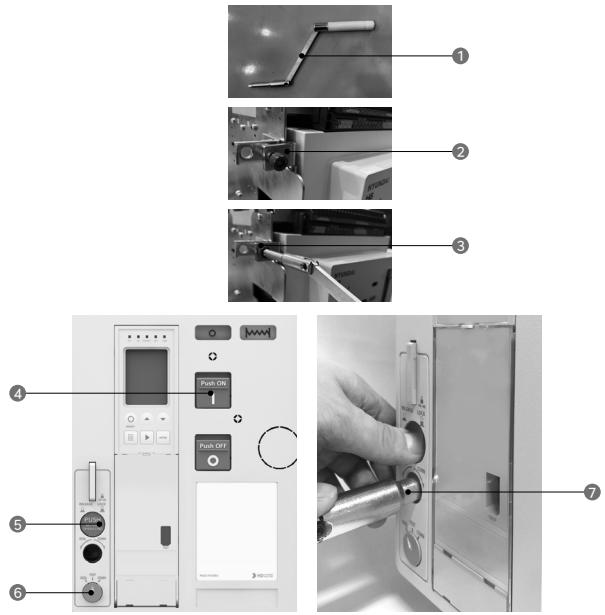
1) Moving from CONNECTED position to TEST position

- When the fixing blocks are fitted, loosen and free right and left screws. (Fig. 1. ③)
- Check that the breaker is open.
  If it is closed, push open button (Fig. 1. ④) to open breaker.
- When the position lock release button is completely pushed in, connect drawin/out handle to shaft fully. (Fig. 1. ⑦) If it is not completely pushed in or not inserted, the body doesn't move and indicator doesn't rotate.
  The position indicator is rotated while the breaker body drawn out.

! CAUTION

When the main circuit is disconnected as the breaker body is being drawn out, the breaker body will be slightly pushed forward by the spring action of the primary disconnect contacts with a "banging" sound.
While the loudness of this sound may vary, the sound itself is perfectly normal and does not affect breaker performance.

Fig.1



! CAUTION

- When position lock release button (Fig. 1. ⑤) is pushed, the breaker will not close. In order to test operation, must move to the right position (CONNECTED, TEST, ISOLATED), and position lock release button (Fig. 1. ⑤) should be pushed out.
- When breaker body is drawn out to the TEST position or ISOLATED position, a metallic sound "click" will be heard, and the draw-in/out handle operation is automatically locked.
  At this time, do not attempt to compulsively rotate the handle.

2) Moving from TEST position to ISOLATED position

In order to move from TEST to ISOLATED, push position lock release button (Fig. 1. ⑤) shortly after handle locking is released, and turn the draw-in/out handle in a counterclockwise direction.

! CAUTION

If it is approaching ISOLATED position, draw-in/out handle (Fig. 1. ①) will be automatically locked.
Do not attempt to compulsively rotate the handle.

3) Removing body from ISOLATED position

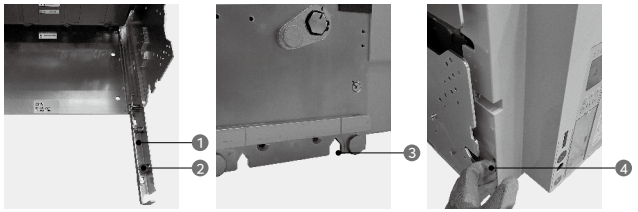
For maintenance, inspection, or exchanging parts, the breaker in the ISOLATED position should be drawn out from the draw-out cradle. If the breaker is charged, the breaker should release the closing springs, by using the manual close and open button.

- To remove the ACB Body from the cradle in the ISOLATED position, grasp both sides of the ACB Body (Fig 2. ④) and pull it out completely.
- Once the breaker is fully extended, slowly pull the breaker out from the side Rails of the Draw-Out Cradle (Fig 2. ①).

! CAUTION

- This operation should be done slowly. Apply wire rope to the lifting lug and lift the body upward after releasing stoppers.
- Don't leave the breaker in REMOVED position.
  When the breaker is drawn out, the center of gravity is changed.
  The draw-out cradle should be fixed in advance in the PNL.

Fig.2



Draw-in Operation

In order to push back the breaker body to CONNECTED position, follow steps below.

- Check that the breaker is open. Also, check draw-in/out indicator shows isolated. Check charged/discharged indicator. Indicator shows discharged「」.
- Pull out the side rails of the Draw-Out Cradle (Fig 2. ①).
- Use a lifter or lift the breaker, and ensure the side guides of the breaker body (Fig 2. ③) align perfectly with the rail guides (Fig 2. ②) on the cradle rail (Fig 2. ①) and push the breaker in completely. Do not push it in forcefully.

CAUTION

Never enter under the breaker. The breaker may fall and cause serious injury. Do not push hard the breaker.

- With position lock release button (Fig. 1. ⑤) pushed, connect drawing in/out handle to the shaft.
- Then, when position lock release button (Fig. 1. ⑤) is pushed, it maintains self holding and allows draw-in/out handle (Fig. 1. ①) operation.
- To move the breaker to CONNECTED or TEST position, rotate draw-in/out handle in a clockwise direction. With the breaker moving, if position indicator is accessing TEST position, position release button (Fig. 1. ⑤) is automatically pushed out and it locked draw-in/out handle. When the handle is locked, do not rotate excessively. This may cause damage to the breaker.
- In the TEST position, with the handle stick, PUSH position release button (Fig. 1. ⑤) again and release draw-in/out handles.
- With position lock release button pushed, it maintains self holding, rotate draw-in/out handle in a clockwise direction again.
- Main circuit connection starts just before CONNECTED position, draw-in/out handle operation will become heavy. But continue the rotating by adding force until, position indicator (Fig. 1. ⑥) of the breaker shows CONNECTED and, position release button (Fig. 1. ⑤) is automatically pushed out to lock the draw-in/out handle. Then, remove draw-in/out handle and fix it on the right side of draw-out cradle.
- In CONNECTED position, do not operate draw-in/out handle in a clockwise direction. This may cause damage to the breaker.
- When the breaker fixing block is fitted, tighten left and right fixing screw by draw-in/out handle.
- In order to control the breaker electrically, you should connect all circuit according to the diagram on page 4.

Periodic Inspection and Parts Replacement

Period of periodic inspection

It is most appropriate that the user works out his or her own inspection plan for the breakers according to the switching frequency, the value of normal breaking and making currents, the magnitude of the fault current interrupted, service conditions, and environmental conditions.
It is recommended to perform a simplified inspection once every 6 months and a full inspection once every 12 months.
To perform an periodic inspection, draw out to ISOLATED position or move out of draw-out cradle.

Mechanical durability life of the breaker

The following table shows the mechanical durability of the breaker for Maximum Warranty Life. When accumulated counting number of switching cycles excees those shown, we recommend you check the breaker.
For renewal or thorought inspection, please contact us.

Frame Size	Up to 2,500 AF	Between 3,200 AF and 4,000 AF	At 6,000 AF
Number of switching cycle	Total : 12,500 (Mechanical : 9,700, Electrical : 800)	Total : 4,000 (Mechanical : 3,200, Electrical : 800)	Total : 1,500 (Mechanical : 1,100, Electrical : 400)

* Electrical endurance tests were performed at the minimum rated current to reflect the most demanding arc interruption conditions.

Frequency of inspection

Frequency of inspection is considered with service condition and switching numbers, breaking/making current value, but it is recommended to perform a simplified inspection once every 6 months and a full inspection once every 12 months.

Inspection by switching numbers

Switching Conditions of ACB	Inspection Interval Based on Switching Cycles		
	Below 1,000 AF	1,250~2,500AF	Above 3,200 AF
Switching operation in the state of nearly no-carrying current	1,000	1,000	1,000
Switching operation in the rated current region	500	500	100
Switching operation in overload region (about 2-3 times the rated current)	25	25	10
Switching operation in current interruption region	each time	each time	each time

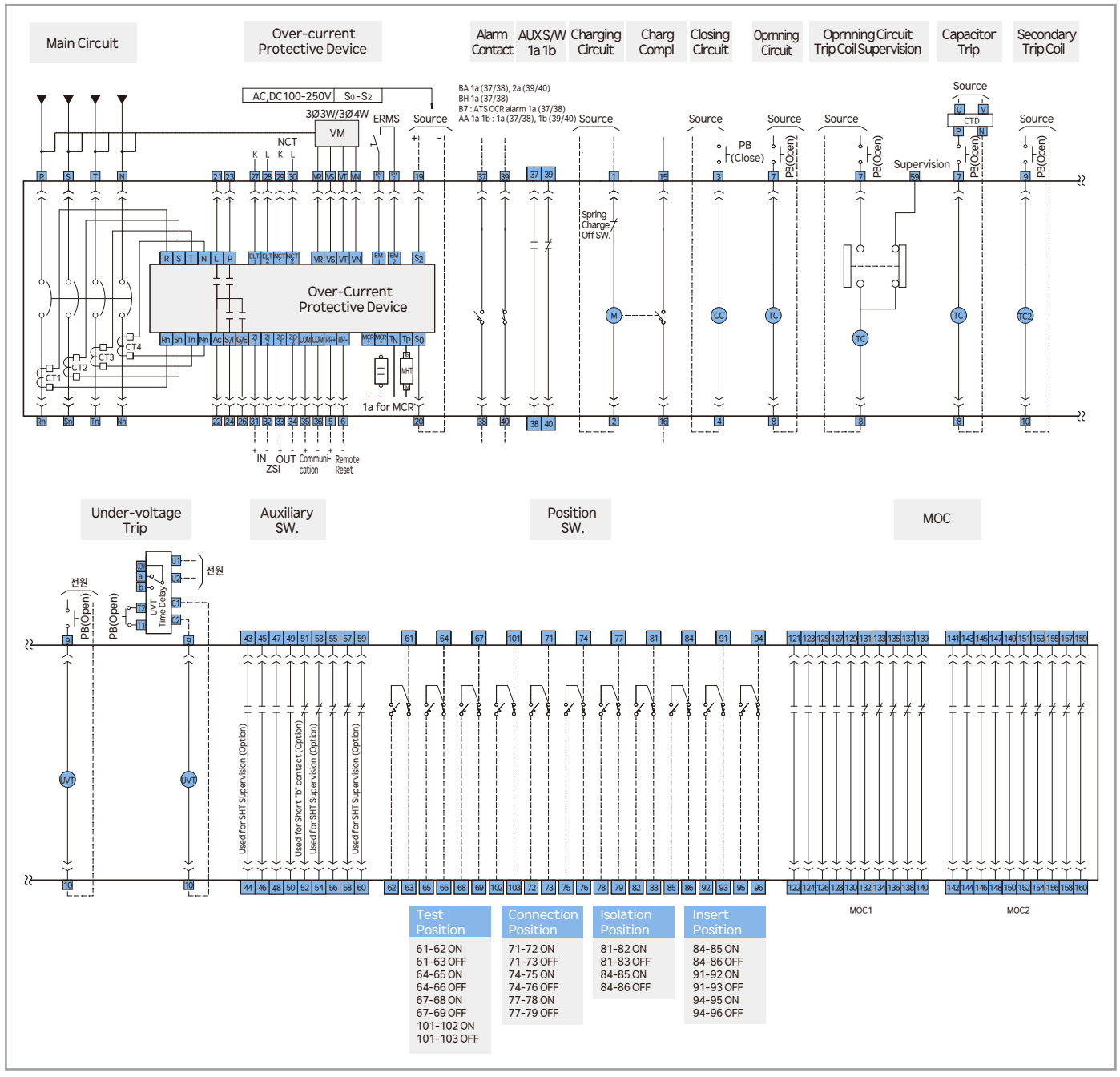
CAUTION

Draw out the breaker to the ISOLATED position or remove the breaker body from the draw-out cradle for inspection or parts replacement purposes. Make sure that residual heat of terminal should be cooled down before performing inspection work. Unless it may cause burn.

Troubleshooting Points

No.	Trouble	Pheno-menon		Expected Causes	Corrective Actions
		Ma-chine	Elec-trical		
1	Close fault	X	X	① Mechanism badness	① Contact HE
				② Whether it is charging by handle mechanism	② Contact HE
				③ Is close rod returned?	③ Close rod returning check
				④ Whether inertia latch is jamming or not?	④ Inertia latch movement check
				⑤ Is ACB placed in proper position (ISO-TEST-CON)?	⑤ Move to proper position
				⑥ Is GPR manual reset button pull out?	⑥ Push back GPR manual reset button
				⑦ Is UVT coil applied proper voltage?	⑦ UVT source check
				⑧ Is position lock button handle cover pushed in?	⑧ Handle cover comeback. Push draw in/out handle and adjust position confirm button returning
				⑨ The breaker caught interlock?	⑨ Interlock dissolution
				⑩ Is trip latch turned down?	⑩ Trip latch returning
		X	O	① Is close rod returned?	① Close rod returning
				② Is trip latch turned down?	② Trip latch returning
		O	X	① Is proper voltage applied to CC coil?	① CC coil voltage check
				② Is CC coil operates normally?	② Coil replacement
				③ Is CC coil stroke in proper position?	③ Stroke check
				④ Is UVT coil applied proper voltage?	④ UVT source check
				⑤ Is over current relay reset?	⑤ Reset OCR
2	Open fault	X	X	① Whether any interlock system exist?	① Contact HE
				② Whether trip latch exists bur?	② Contact HE
		O	X	① Whether proper voltage applied to TC coil?	① TC coil voltage check
				② Whether TC coil exists mechanical interference	② After voltage is applied, check core movement
				③ Whether proper voltage applied to UVT coil?	③ UVT coil voltage check
3	Charging fault	X	X	① Whether handle mechanism cam operates normally?	① Contact HE
		O	X	① Whether proper voltage applied to motor?	① Voltage check (85~110% of rated voltage)
				② Whether motor control unit is damaged	② Motor output voltage check
				③ Whether motor ratchet (M/B) is proper	③ Part check

Wiring Circuit for ACB



Symbol description

CT	Current transformer
L	LTD terminal
PT	PTA (pre-trip alarm) terminal
G	Ground fault terminal
S / I	STD/INST terminal
Ac	Common terminal
NCT	Neutral current transformer
ZI	Zone selective input
ZO	Zone selective output
MCR +,-	MCR input terminal
Tp / Tn	MHT input source
M	Charging motor
CC	Close coil (close)
TC	Trip coil (open)
UVT	Under voltade trip coil
CT	Magnetic hold trigger
S0 / S2	GPR protection relay source power
EM1 / EM2	ERMS

- RR / Remote Reset
- VM / Voltage Module
- VR~VN / Voltage phase input
- R~N / Current input
- Rn~Nn / Current input

Terminal description

1 2	Charging motor (M) source power
3 4	Closing coil (CC) source power
7 8	Open coil (TC) source power
9 10	UVT coil terminal
15 16	Charge complete contact
19 20	OCR source power
22 21	LTD contact
22 23	PTA/TEMP contact
22 24	STD/INST contact
22 26	GFT/ELT contact
22 30	NCT input terminal
31 ~ 34	ZSI
41 ~ 60	AUX. contact
61 ~ 93	Position switch

- Manufacturer's wiring
- User's wiring
- Disconnecting device (draw out type)

Control Jack Lay-out

OCR & OCGR																CONTROL CIRCUIT					AUXILIARY SW.									
POW	OCR CONTACT				RR	NCT	ZSI-IN	ZSI-OUT	ERMS	RS485	BA(BH)/AA		VM/ZCT		M	CC	TV/TCS	UVT/TC2	CHA	1a	2a	3a	4a	1b	2b	3b	4b	5b		
	+19	21	23		5	+29	+31	+33	EM1	+35	37	39	VR/27	VT	1	3	7	9	15	43	45	47	49	51	53	55	57	59		
-20	22	24	26	6	-30	-32	-34	EM2	-36	38	40	VS/28	VN	2	4	8	10	16	44	46	48	50	52	54	56	58	60			
OCR Protection relay															Operating circuit					Auxiliary switch										

OCR & OCGR														
POW	OCR CONTACT				RR	NCT	ZSI-IN	ZSI-OUT	ERMS	RS485	BA(BH)/AA		VM/ZCT	
	+19	21	23		5	+29	+31	+33	EM1	+35	37	39	VR/27	VT
-20	22	24	26	6	-30	-32	-34	EM2	-36	38	40	VS/28	VN	

- GPR Protection relay -

CONTROL CIRCUIT				
M	CC	TV/TCS	UVT/TC2	CHA
1	3	7	9	15
2	4	8	10	16

- Operating circuit -

AUXILIARY SW.									
1a	2a	3a	4a	1b	2b	3b	4b	5b	
43	45	47	49	51	53	55	57	59	
44	46	48	50	52	54	56	58	60	

- Auxiliary switch -